

Open Source Testing Solutions that Make Software Reliable



This article lists a few open source testing tools that can help developers validate their software.

With the arrival of the agile and DevOps practices, the software development industry is undergoing major disruption. The reliability and stability of software has become critical, increasing the importance of software QA. Organisations adopt DevOps practices and establish CI/CD pipelines to enable delivery of small chunks of software in fast and frequent iterations. QA teams are moving towards the 'shift left' approach, enabling early testing of the software. These test teams need to have a result-oriented collaboration with the development teams, who are also involved in the QA process. Development teams create and run unit tests, while the test teams validate these at the API and UI layers.

More and more application development is moving towards cloud native technologies. Applications are being built with service packages and containers, deployed as microservices, and managed on cloud elastic infrastructure through DevOps processes and continuous delivery workflows.

Test automation helps to achieve a larger test coverage and higher level of regression testing to ensure the stability of the software.

Performance engineering is replacing performance-testing techniques. The focus in the former is on analysing how to prevent and forecast software performance issues.

Open source testing tools

Test tools need to be selected depending on the type of applications

and interfaces they are exposed to. Their coverage should encompass all test areas, be it functional or non-functional testing. Functional test areas include Web UI, mobile app UI, and API validation. Non-functional test areas include performance, reliability, scalability and accessibility.

There is a range of open source tools that can help to validate the software in all these areas. Let's take a quick look at some of these.

BDD based test automation framework testing tools

The BDD or Behaviour Driven Development framework is a software development approach that allows the tester or business analyst to create test cases in simple text language (English), so that even non-technical team members can understand what is going on in the software project. A few popular BDD based test frameworks are discussed below.